

Climate-resilient grapevine diversity

Spatial scale: vineyard

About this sustainable management strategy

Managing grapevine diversity is a long-term strategy that strengthens vineyard resilience by making use of the wide range of characteristics found in grape varieties and rootstocks. Differences in growth cycle, tolerance to drought, heat, frost, and diseases, as well as rooting behaviour and water uptake, allow planting material to be matched more closely to local environmental conditions and production objectives.

Vineyard renewal provides an opportunity to reassess whether existing planting material remains suitable for individual vineyard blocks. Rather than relying on a single variety–rootstock combination, growers can introduce a broader range of locally adapted cultivars, disease-resistant varieties, or drought-tolerant rootstocks according to soil characteristics, water availability, elevation, and exposure. This approach helps spread production risks, increases the adaptive capacity of vineyard ecosystems, and can reduce the need for external inputs such as plant protection products.

Maintaining and using grapevine genetic diversity also contributes to the conservation of locally adapted varieties and regional viticultural heritage. Combined with other ecosystem-based management practices, a diverse vineyard landscape is better equipped to respond to changing environmental conditions while sustaining grape quality and long-term productivity.

Key ecosystem services

- Agrobiodiversity conservation (*maintenance and use of grapevine genetic diversity*)
- Sustainable grape production (*greater production stability under climate variability*)
- Reduced plant protection requirements
- Cultural heritage (*conservation of traditional and locally adapted grape varieties*)

Considerations for successful implementation

Selection according to local site conditions

The choice of varieties and rootstocks should consider soil characteristics, water availability, topography, climatic conditions, and long-term climate projections. Planting material that performs well in one location may not be suitable for another.

Balance climate resilience, market requirements and production regulations

Adaptation strategies should consider grape quality, yield potential, wine style, and market requirements alongside climate resilience. The suitability of new or alternative varieties should be evaluated within the context of local production systems and appellation regulations, where applicable.

Long-term planning

Changing varieties or rootstocks represents a long-term investment with implications for vineyard management over several decades. Decisions should be based on future climate scenarios, vineyard objectives, and the expected lifespan of the planting, while considering economic feasibility and the availability of suitable planting material.